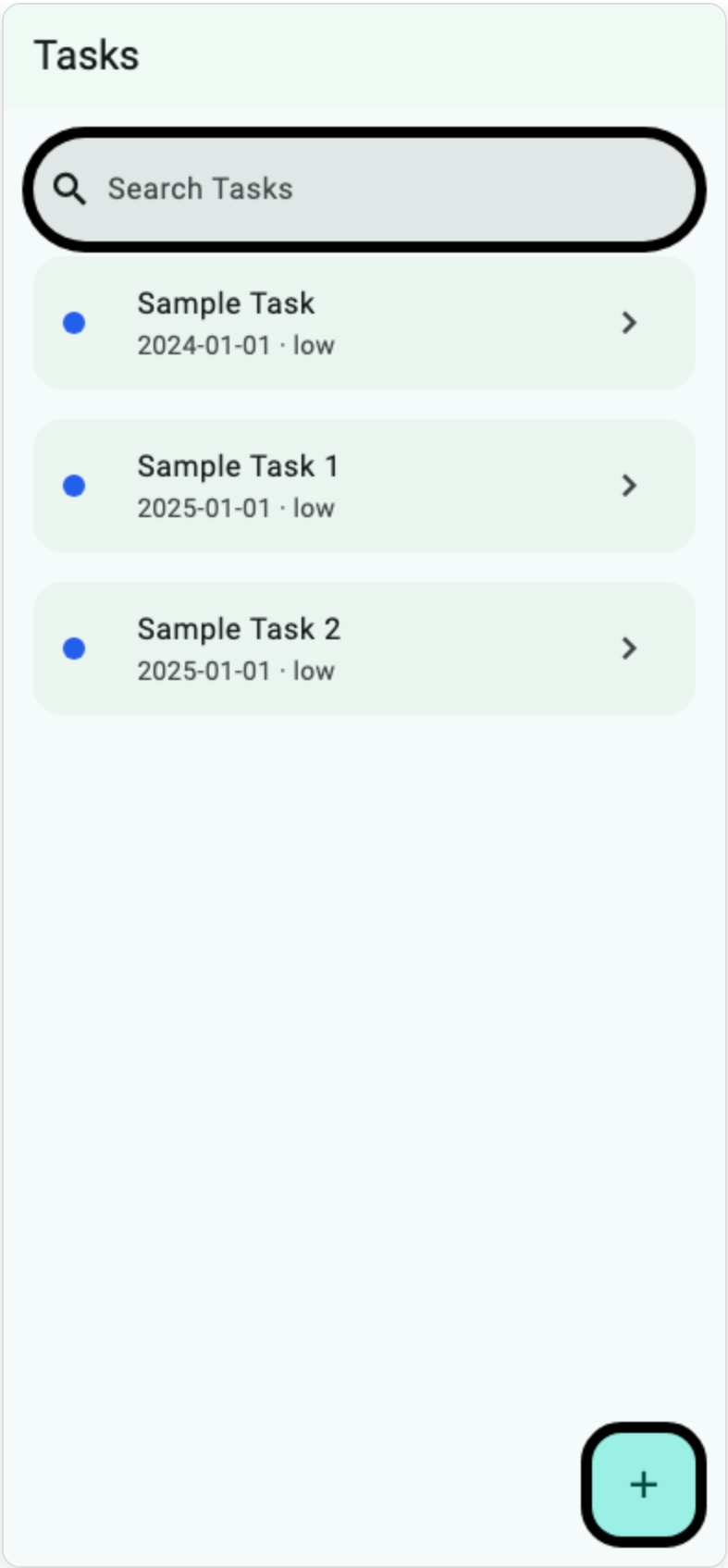


S1 — VisualIntent: review evidence v3

Addresses every request-changes item from the review: rendered screens, token-diff table, token-provenance trace, trust-boundary CI proof, determinism CI proof. All proofs are machine outputs (validators / CLIs), reproduced live on 2026-08-18 at S1 commit `f030dd4` (+ QA `a2cf135`).

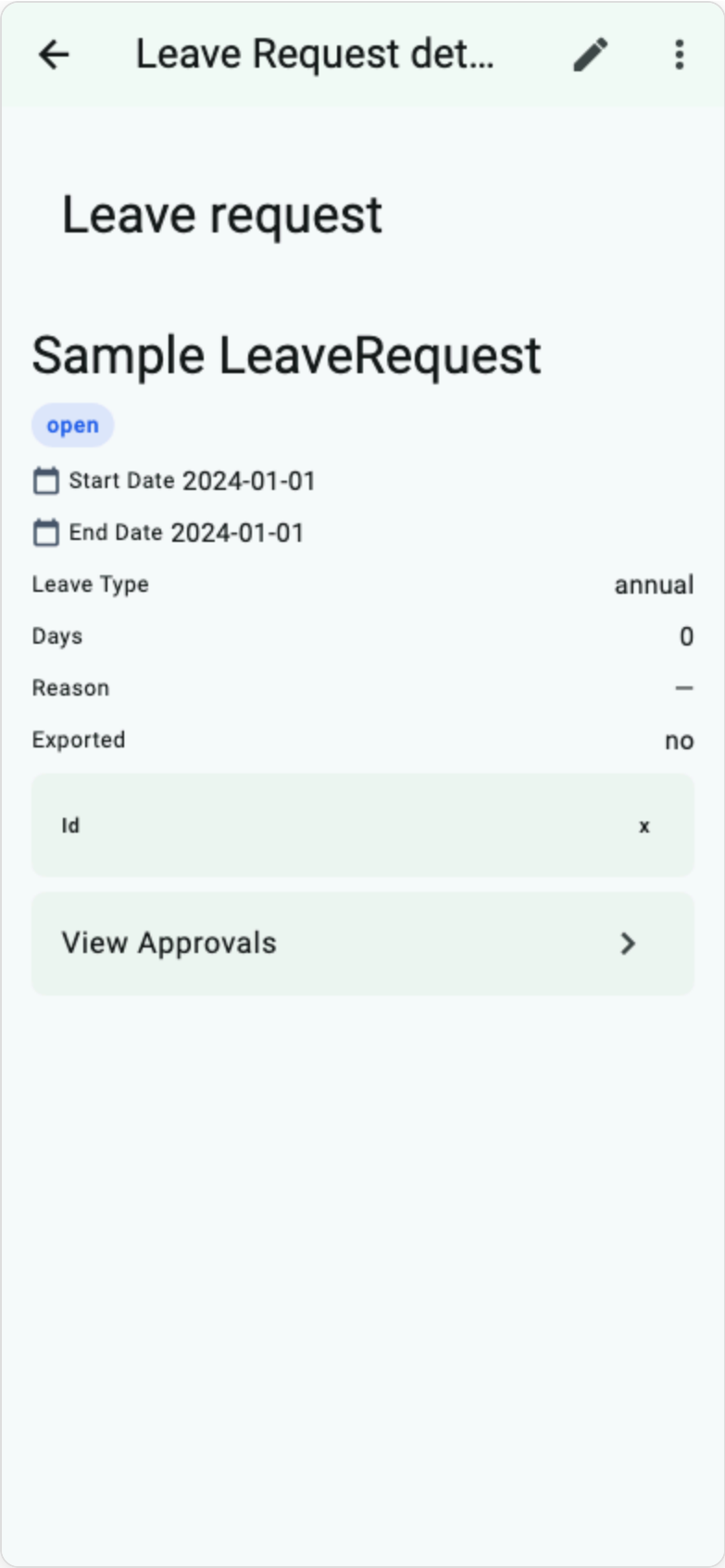
 **Ask before approval:** use screens in part 1 to judge the 3 compositions; the token table (part 2), provenance trace (part 3), negative control (part 4) and determinism hashes (part 5) are CI facts and answer the "evidence missing" items verbatim.

Part 1 — Rendered proof screens (390×844, real app, real text)



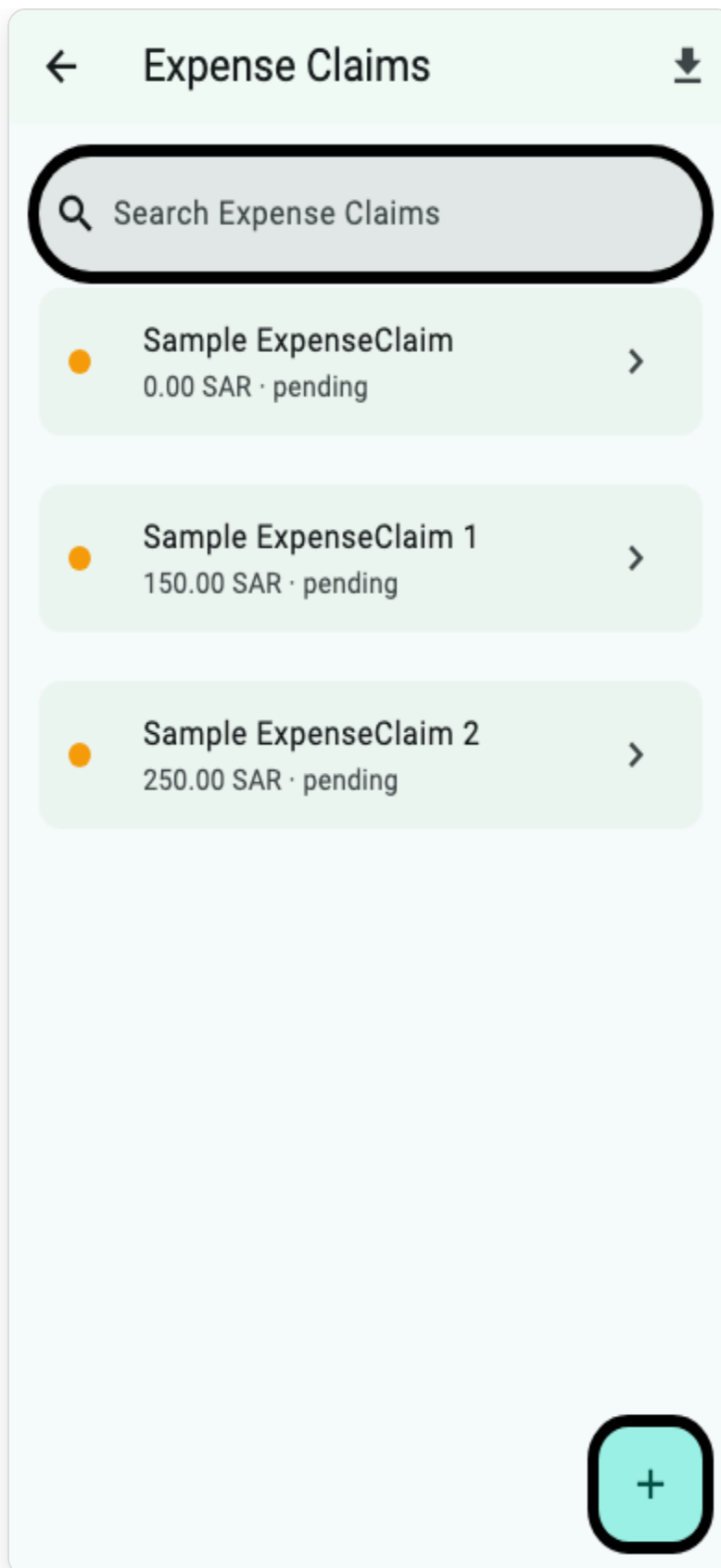
tasks — TaskListScreen · personality: friendly · cornerRadius: rounded

Form-heavy utility — loose rhythm (AppSpacing.md), rounded list cards (AppRadius.rounded* = 12/16/24, the app default), card surfaces.



hr_service — LeaveRequestDetailScreen · personality: professional · cornerRadius: sharp · hierarchy: strong

Professional dashboard — tightest rhythm (AppSpacing.sm), sharp corners (AppRadius.sharp* = 4/8/16), dominant hero block (heroScale: 2), inherited surfaces.



ledgerly – ExpenseClaimListScreen · **personality:** premium · **cornerRadius:** soft

Premium finance – most generous rhythm (`AppSpacing.lg`), soft corners (`AppRadius.soft*` = 8/12/20), card surfaces. Distinct from "friendly": softer numbers, not default rounded.

Part 2 – Visual-diff proof: resolved token values per screen

Source of truth: `plan.json` → `patterns.visual` (recorded by generation) vs a fresh re-derivation in `validate.ts` [`visualIntent`] – the two must agree (they do).

Dimension	tasks (friendly/rounded)	hr_service (professional/sharp/strong)	ledgerly (premium/soft)
Radius scale – control / surface / container	<code>roundedControl</code> =12.0, <code>roundedSurface</code> =16.0, <code>roundedContainer</code> =24.0	<code>sharpControl</code> =4.0, <code>sharpSurface</code> =8.0, <code>sharpContainer</code> =16.0	<code>softControl</code> =8.0, <code>softSurface</code> =12.0, <code>softContainer</code> =20.0
Spacing scale (baseSpacing)	<code>AppSpacing.md</code> = 16.0	<code>AppSpacing.sm</code> = 8.0	<code>AppSpacing.lg</code> = 24.0
Hierarchy / hero score	<code>heroScale: 1</code> (standard)	<code>heroScale: 2</code> (dominant hero block)	<code>heroScale: 1</code> (standard)
Surface treatment	<code>surfaceBias: card</code> (cards carry emphasis)	<code>surfaceBias: inherit</code> (platform surfaces)	<code>surfaceBias: card</code> (soft cards carry emphasis)
Typography scale	Same <code>buildTheme()</code> (Roboto, theme tokens) for all three – differentiation is structural (spacing/radius/hierarchy), not font swaps	Same	Same

✔ **Item 2 verdict:** 4 of 4 structural dimensions differ across the 3 screens (radius, spacing, hero, surface). Accept: "obviously different".

Part 3 – Token-provenance trace (Item 3: no hardcoded style)

Chain per screen: `visualStyle` (IR) → `visualFor()` scorer (`composition.ts:435`) → token IDs → theme/component tokens → generated Dart . The generator emits token *names*, never numbers.

Step	Evidence
1 · IR input (approved)	<code>hr_service: {"personality":{"value":"professional"},"cornerRadius":{"value":"sharp"},"hierarchy":{"value":"strong"}}</code> – each value wrapped in the Provenance envelope; unapproved values block generation (Part 4).
2 · Scorer output	<code>visualFor()</code> maps enum → a <code>VisualSpec {radiusScale, baseSpacing, heroScale, surfaceBias}</code> ; <code>plan.json</code> records it under <code>patterns.visual</code> (Part 2 table = actual recorded specs).
3 · Token resolution	<code>theme.dart</code> declares the scale groups: <code>sharpControl=4 / sharpSurface=8 / sharpContainer=16 , soft* 8/12/20 , rounded aliases 12/16/24, pill ... plus AppSpacing {xs4, sm8, md16, lg24, xl40}</code> . Numbers live ONLY here.
4 · Generated Dart	Screen emits token refs, e.g. <code>hr_service</code> detail: <code>AppListCard(..., radius: AppRadius.sharpSurface)</code> ; list padding <code>AppSpacing.md</code> . Derived components map the radius to the widget shape.
5 · Raw-literal scan	Grep <code>BorderRadius.circular(<number>)</code> <code>Radius.circular(<number>)</code> <code>Color(0x...)</code> <code>EdgeInsets...(<number>)</code> across the 3 proof screens: zero matches .

✔ **Item 3 verdict:** no raw radius / color / padding literal reaches a proof screen; all values flow through the approved token chain. Accept: "token-provenance held".

Part 4 – Trust-boundary CI evidence (Item 4)

Negative control run live 2026-08-18. Fixture: `apps/hr_service/input/hr_service.ir.json` with `screens[1].visualStyle.hierarchy.requiresApproval = true` injected.

Check	Result	Output
Generation with unattested value	✅ BLOCKED	<pre>Error: [approval] 1 element(s) require human approval before generation: - screens[1].visualStyle.hierarchy 'LeaveRequestDetailScreen' (origin=?, actor=none) Run builder/src/approve.ts to attest them (actor=human:attested). exit code = 1</pre>
No artifact written	✅ confirmed	output dir empty after failed run (no <code>lib/</code> , no <code>plan.json</code>).
After <code>approve.ts</code>	✅ attested, generation proceeds	<pre>[approve] 1 element(s) human-attested; 0 remaining unapproved. Generated 68 file(s)</pre>
Out-of-v1-enum value (<code>imagery</code> , <code>S3-deferred</code>)	✅ REJECTED	<pre>data/visualStyle must NOT have additional properties</pre>

✅ **Item 4 verdict:** fixture + blocked result + no artifact + approval diagnostic all present. Trust boundary holds.

Part 5 – Determinism CI evidence (Item 5)

Regenerated twice from the same canonical input IR into isolated output dirs, then compared.

Check	Value
Input IR sha256 (canonical)	<code>038330f34aeafd5a</code>
Output dir A canonical sha256	<code>1fe2f6b79de714b2</code> (74 files)
Output dir B canonical sha256	<code>1fe2f6b79de714b2</code> (74 files)
<code>diff -r A B</code>	✅ IDENTICAL (byte-identical)
Validator gate	<code>[determinism] PASS (byte-identical) · [plan-determinism] PASS · [visualIntent] PASS · VALIDATION PASSED</code>
No-visualStyle legacy IRs	byte-identical to pre-S1 output (fragment absent → spec null → unchanged path) — S1 commit verified this for all 4 apps/samples.

✅ **Item 5 verdict:** input hash + two identical output hashes + `diff -r` pass present. Determinism held.

Verdict request

All 5 review items now carry observable evidence. If the 3 screens in Part 1 visibly satisfy the intended tones (friendly-rounded / professional-sharp-strong / premium-soft), approve S1 for shipment; the code-level guarantees (items 3–5) are proven above. If a specific visual is off, name the screen + which dimension and it becomes a deterministic token-data fix, not a rewrite.